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Another important benefit of Green Building is Energy efficiency which results in reduced energy consumption for AC and Heating needs. Effective use of natural lighting, cool roof and wall panels which leads directly cost savings to the building owners.

The use of Solar Energy in green building can provide free electricity for the building owners. Solar panels can be installed on the roof top of the building where the solar energy will be converted to electricity.

What is Green Building?A green building is one which uses less water, optimizes energy efficiency, conserves natural resources, generates less waste and provides healthier spaces for occupants, as compared to a conventional building What is Green Building?A green building is one which uses less water, optimizes energy efficiency, conserves natural resources, generates less waste and provides healthier spaces for occupants, as compared to a conventional building E- WASTE What is e-waste and why is it a problem?

The discarded electronic and electrical equipment is called e-waste. They are harmful to the environment because of the presence of harmful chemicals like brominated flame retardants and toxic heavy metals like mercury, cadmium, lead etc.

What are examples of e-waste?

Mobile phones, television, refrigerator, home appliances etc. are some of the examples of the e-waste.

How is e-waste managed?

There are national e-waste policies for e-waste management. These policies help the e-waste collected and recycled.

Also, such provisions hold stakeholders responsible for e-waste pollution. In India, there are e-waste management rules 2016 that guide the e-waste management.

What are the ten categories of e-waste?
The types of e-waste are listed down below:

- Major Appliances
- Small Appliances
- Computer and Telecommunications Appliances
- Lighting Devices
- Electrical and electronic tools
- Electronic Toys
- Medical devices
- Monitoring Devices
- Vending Machines

What is the environmental impact of e-waste?
The toxic chemicals present in the e-waste releases into the atmosphere when it warms up. The lead, zinc, barium
etc, found in electronic waste tend to harm human health. Also, when e-waste meets the landfills, the toxic chemicals
seep into the soil and water, leading to groundwater pollution that further impacts sea health and human health.
POLLUTANTS IN E-WASTE:
Pollutants or toxins in e-waste are typically concentrated in circuit boards, batteries, plastics, and
LCDs (liquid crystal displays). Pollutants and their occurrence in waste electrical and electronic
equipment are:

POLLUTANTS	OCCURRENCE
Arsenic	Semiconductors, diodes, microwaves, LEDs (Light-emitting diodes), solar cells
Barium	Electron tubes, filler for plastic and rubber, lubricant additive
Brominated flame- proofing agent	Casing, circuit boards (plastic), cables and PVC cables
Cadmium	Batteries, pigments, solder, alloys, circuit boards, computer batteries, monitor cathode ray tubes (CRTs)
Cobalt	Insulators
Copper	Conducted in cables, copper ribbons, coils, circuitry, pigments
Lead	Lead rechargeable batteries, solar, transistors, lithium batteries, PVC (polyvinyl chloride) stabilizers, lasers, LEDs, thermoelectric elements, circuit boards
Liquid crystal	Displays
Lithium	Mobile telephones, photographic equipment, video equipment (batteries)
Mercury	Components in copper machines and steam irons; batteries in clocks and pocket calculators, switches, LCDs
Nickel	Alloys, batteries, relays, semiconductors, pigments

PCBs (polychlorinated biphenyls) Transformers, capacitors, softening

agents for paint, glue, plastic

E-waste Management Rules in India
The Environment, Forest, and Climate Change Ministry (MoEF&CC) have announced the E-Waste
Management Rules 2016. These new rules replaced the earlier E-Waste (Management and Handling) Rules
of 2011.
The new Rules make for stricter norms and are a part of the government's increased commitment towards
environmental governance.
Highlights of the new E-waste Management Rules 2016:
1. It includes CFLs or Compact Fluorescent Lamps as well as other lamps with mercury, and similar
equipment.
2. The Rules for the first time, bring producers under the ambit of the Extended Producer
Responsibility or EPR, together with the targets.
3. Producers have been made accountable for e-waste collection and e-waste exchange as well.
4. Additional stakeholders included are:
1. Manufacturers
2. Dealers
3. Refurbishers and Producer Responsibility Organizations.
5. Compact Fluorescent Lamp (CFL) and other mercury-containing lamps have been brought under
the purview of the rules.
India's Environment Ministry has notified rules targeting the wide range of groups like hotels, residential
colonies, bulk producers of consumer goods, ports, railway stations, airports, and pilgrimage spots. This is
to ensure that the solid waste generated in their facilities is treated and recycled.
Key Points of E-waste Management Rules 2016:
Local bodies with a population of one lakh or above were supposed to establish solid waste
processing facilities within two years,
Census towns below a lakh would be given three years to establish solid waste processing facilities
Old and discarded dump sites would have to be shut-down or bio-remedied within five years.
The rules on solid waste management have been amended after 16 years.
Garbage management is the responsibility of municipal bodies, they would have the rights to charge
user fees and levy spot fines for littering and non-segregation.
A transition period of two to five years would be in place beyond which fines would be imposed as
per the country's Environment Minister.

What is Biomedical Waste?

Biomedical waste is any waste containing infectious or potentially infectious materials. These wastes are generated during the diagnosis, treatment, and immunization of humans and animals. Biomedical wastes can be in both solid and liquid forms. Examples of biomedical wastes include:

- Waste sharps such as needles, lancets, syringes, scalpels, and broken glass
- Human tissues or identifiable body parts (as a result of amputation)
- Animal tissues and waste from veterinary hospitals
- Used bandage, dressings, gloves, and other medical supplies
- Liquid waste from infected areas
- Laboratory wastes

Biomedical wastes are distinct from regular garbage and require particular disposal and treatment.

Types of Biomedical Waste

The World Health Organization (WHO) has categorized biomedical waste into eight categories. They are:

1. Infectious Waste - Any biomedical waste that is infectious or contaminated.
2. Sharps - Sharps objects like needles, scalpels, broken glass, and razors.
3. Pathological Waste - Body parts of humans or animals, including tissues, fluids, or blood.
4. Pharmaceutical Waste - Unused drugs, medicine, or creams that are expiring.
5. Genotoxic Waste - Toxic drugs and hazardous toxic waste
6. Radioactive Waste - Any waste containing potentially radioactive materials
7. Chemical Waste - Liquid waste from machines, batteries, and disinfectants is chemical.
8. General/Other Waste - All other non-hazardous waste.

Further, the Central Pollution Control Board (CPBC) has designated separate colour-coded bins to dispose of biomedical wastes as per their nature.

1. Yellow Bin: For anatomical waste, chemical waste, soiled waste, chemotherapy waste, discarded linen and medicines, and laboratory waste.
2. Red Bin: For contaminated plastic wastes
3. Blue Bin: For glass waste and metallic implants
4. Black Bin: For hazardous and other waste

The wastes in each of the bins have different treatment and disposal methods.

There are similar topics important from the environment and ecology point of view. Complement your reading about the environment and air with these topics:

Effects of Biomedical Waste

Exposure to hazardous biomedical waste can cause disease or injury to human health. HIV, hepatitis B, and C are the three most commonly spread viruses worldwide due to improper treatment of medical wastes.

They are transmitted through injuries from contaminated syringes and needles.

Doctors, nurses, and sanitation workers are amongst the most vulnerable to the harmful effects of

biomedical waste.

At a time of rapid emergence of new strains of the novel coronavirus, the importance of appropriate
treatment of medical wastes cannot be more emphasized. The various technologies that can be used for
treatment include:

- Incineration
- Chemical Disinfection
- Wet Thermal Treatment
- Microwave Irradiation
- Land Disposal
- Inertization

ENVIORNMENTAL ETHICS: ISSUES AND POSSIBLE SOLUITONS:

Introduction:

Environmental ethics is the branch of philosophy which analyzing about human use of Earth′s limited resources. Environmental ethics refers to the issues, principles, and guidelines
relating to human interrelations with their environment.

Human Centric Thinking:

If we think, ‘Man is all powerful and the supreme is the master of nature and can harness it at his will′, it
reflects our human - centric thinking.

Earth - Centric Thinking:

On the other hand, if we think, ‘nature has provided us with all the resources for leading a beautiful life
and she nourishes us like another, we should respect her and nurture her′, this is an ‘Earth Centric
Thinking′.

Environmental Ethics:

Some important ethical guidelines known as ‘Environmental Ethics′ (or)‘Earth Ethics′. Environmental
ethics are as follows. The Environmental Ethics(or) Dharma of Ecology teaches us how to treat the
nature. It includes Cultures,and Traditions that influenced human relation with nature.

- Protection of Renewable Natural Resources.
- Conservation of Non - Renewable Resources.
- Avoid Over Consumerism.
- Adopt sustainability as a way of life.
- Joining Environment and Economics to eradicate poverty.
- Link Environment with Peace and Security

There is an urgent need to treat the entire world as ‘Global Family′ (Vasudika Kutumbam).

o You should love and honor the earth since is has blessed you with life and governs your survival.
o You should be grateful to the plants and animals, which nourish you by giving you food.
o You should not waste your resources on destructive weapons.

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